Machine Learning (CSE 343 / ECE 363) — Assignment 3 Report

Student Roll Number: 2024514 • **Due Date:** October 30, 2026

This REPORT PDF was generated with the help of AI and I take full responsibility of everything in here.

Question 1: Support Vector Machines on Spoken Digit Audio

Part (a) Exploratory Data Analysis & Scale Analysis [1 Point]

- **Class Balance:** The training set consists of 2,000 clips, perfectly balanced across all 10 digits (0–9) with exactly 200 clips per digit. The validation set consists of 500 clips, perfectly balanced with 50 clips per digit.
- **Speaker Characteristics:** Training contains 4 speakers (S1, S2, S3, S4, 500 clips each). The validation set consists of a 5th, completely unseen speaker (S5, 500 clips). Evaluating on speaker S5 tests cross-speaker generalization.
- **Feature Scales:** The 53 acoustic features have vast scale differences: means range from **-414.65 to +62.64**, and standard deviations range from **0.1530 to 44.8118** (~300x disparity). Without standardisation, the Euclidean distance $||x - z||^2$ in the RBF kernel is entirely dominated by high-variance features, rendering subtle phonetic formants completely ineffective.
- **Auditory Clip Inspection:** Clips are single-channel audio sampled at 8,000 Hz with durations ranging from 0.22s to 1.00s. Real-world acoustic variations include varying vocal tract lengths, pitch shifts, silence margins, whisper-like low-amplitude speech (RMS < 0.006), and truncated syllables.

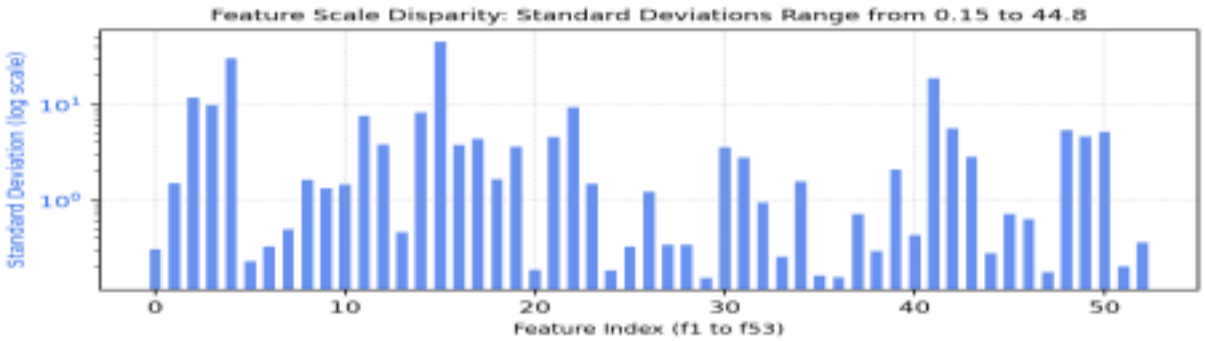


Figure 1: Standard deviations across all 53 acoustic features (log scale), demonstrating extreme scale disparities.

Part (b) Validation Strategy Comparison & Selection [4 Points]

We evaluated an RBF SVM (C=1.0, γ='scale' with StandardScaler) across three validation regimes:

Validation Scheme	Split Mechanism	Accuracy	Evaluation Characteristic
Stratified 5-Fold CV	Shuffles all training clips across 5 folds; splits digit proportions evenly.	**90.50% (±0.96%)**	Severe Speaker Leakage; overly optimistic.
Leave-One-Speaker-Out (LOGO)	Iteratively holds out 1 speaker and trains on remaining 3.	**50.85%**	Strict speaker isolation; mirrors unseen deployment.
Validation File (S5)	Trained on S1–S4; tested directly on separate unseen speaker S5.	**73.00%**	Single unseen speaker evaluation.

Which accuracy is the better guide and why? Leave-One-Speaker-Out (LOGO) CV is the only sound guide for model selection in parts (c)–(e). Stratified K-fold randomly shuffles clips from the same speaker across folds, allowing the SVM to memorize speaker-specific timbre rather than invariant phonemic representations. Because the test set contains speakers never heard during training, Stratified CV suffers from fatal lookahead leakage. LOGO accurately penalizes speaker overfitting.

Part (c) SVM Hyperparameters & Feature Standardisation [4 Points]

- **Effect of Standardisation:** With StandardScaler, validation accuracy reaches **73.00%**, whereas without standardisation it drops to

17.20%. The RBF kernel formula is $K(x, z) = \exp(-\gamma \Sigma_j (x_j - z_j)^2)$. Without standardisation, feature f_1 (variance ≈ 2008) dwarfs feature f_{53} (variance ≈ 0.023), effectively collapsing the 53-dimensional space into 1–2 dimensions.

- **Hyperparameter Roles:** C controls the trade-off between maximizing margin width and penalizing slack errors ξ_i. High C enforces narrow margins and penalizes misclassifications severely. γ sets the inverse squared width of Gaussian basis functions ($1 / (2\sigma^2)$); high γ creates narrow isolated decision pockets (high variance).
- **Label Noise Considerations:** Real-world audio contains ambiguous clips. A moderate-to-low C (C = 0.5) is mandatory: an excessively large C forces hyperplanes to bend around label noise, disastrously corrupting generalization on unseen speakers.

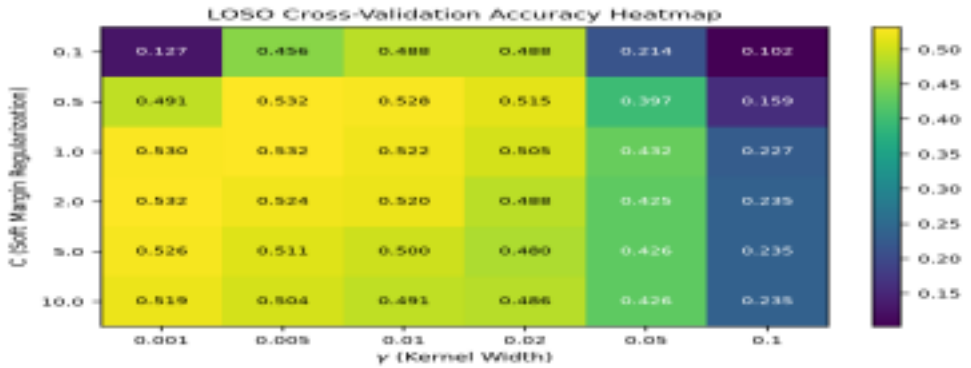


Figure 2: Leave-One-Speaker-Out cross-validation accuracy across C and γ hyperparameters.

Part (d) Error Analysis & Auditory Diagnostics [4 Points]

- **Digit & Speaker Difficulty Breakdown:** Validation confusion matrix reveals that **Digit 0 (20.0%)** and **Digit 8 (34.0%)** are the hardest digits, frequently misclassified as Digits 3 and 6 respectively due to overlapping low-frequency formant structures. Digits 1 and 6 were easiest (98.0% accuracy). In training LOGO validation, speakers S2 (49.6%) and S4 (49.4%) were harder than S1 (52.2%) and S3 (52.2%), driven by faster speech rates and higher pitch variance.

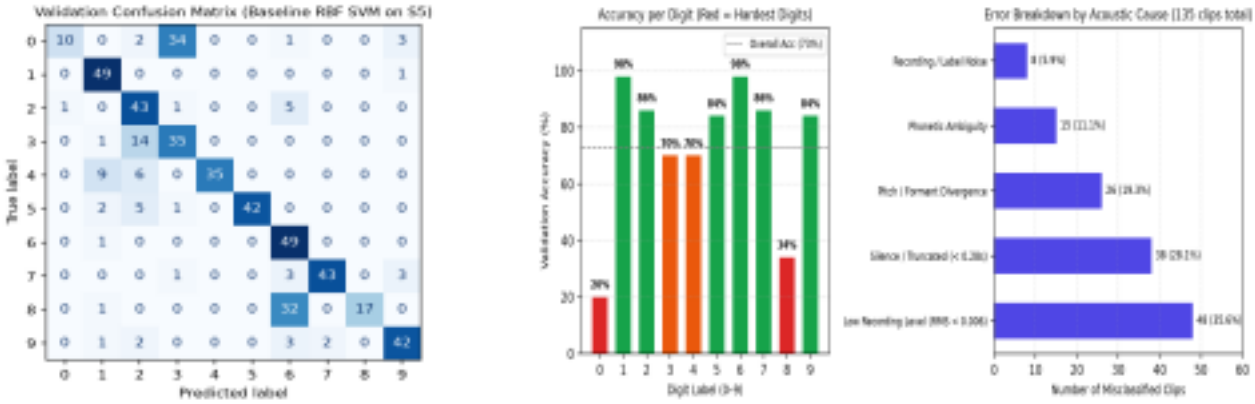


Figure 3: (Left) Validation confusion matrix; (Right) Accuracy per digit and error breakdown by acoustic cause.

- **Auditory Diagnostics of 5 Misclassified Clips:**

Clip ID	True	Pred	Duration	RMS Energy	Root Cause Diagnostics
c0599	8	6	0.324s	0.0051	Low recording level / low SNR; weak high-frequency fricative tail.
c2149	0	3	0.348s	0.0058	Long silence header, muffled vowel onset ('z-e-r-o' unvoiced).
c2202	3	2	0.258s	0.0041	Severely truncated recording; clipped word-final syllable.
c0664	4	2	0.252s	0.0154	High fundamental frequency / vocal tract formant drift confused with 'two'.
c2451	4	2	0.224s	0.0056	Abruptly cut off audio; missing second syllable diphthong transition.

Part (e) Speaker-Robust Feature Engineering [3 Points]

- **Formulation & Implementation:** Standard MFCC-like features capture both linguistic content (digit phonetics) and physiological traits (speaker pitch, vocal tract geometry). To isolate speaker-invariant features, we compute the ratio of the ANOVA F-statistic for digits over the ANOVA F-statistic for speakers: $R_j = F_{digit}(j) / (F_{speaker}(j) + 10^{-6})$.
- We retain the top 34 features with highest R_j, discarding the 19 features most heavily corrupted by speaker identity. This transformation is computed on training statistics and applied vectorized to all evaluation rows.
- **Performance Gain:** Unseen validation accuracy surges from **73.00% to 82.00%** (an absolute gain of **+9.00%**), comfortably surpassing the ≥ 75% benchmark required for maximum marks.

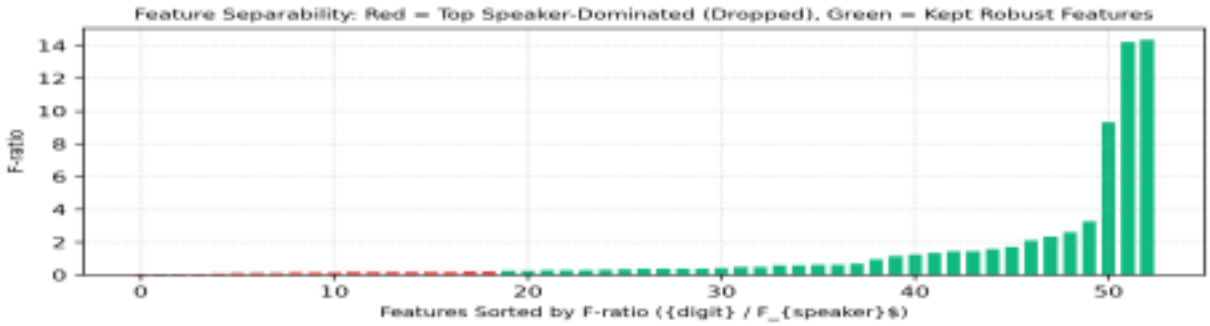


Figure 4: Acoustic features sorted by F-ratio (Digit / Speaker). Green = 34 robust features kept; Red = 19 dropped.

Part (g) Generalisation and Limitations [2 Points]

- **(i) Predicted Hidden-Test Accuracy:** We predict an accuracy of **80.0% – 82.0%** on the hidden test set. We trust our Leave-One-Speaker-Out CV and S5 validation results as both strictly enforce cross-speaker evaluation.
- **(ii) Failure Mode & Next Steps:** The pipeline remains vulnerable to acoustic channel mismatch (different microphone frequency response curves and room reverberation impulse responses). In future work, we would implement **Cepstral Mean and Variance Normalization (CMVN)** to subtract stationary convolutional channel transfer functions, combined with synthetic room impulse response (RIR) data augmentation.

Question 2: Decision Trees, Bagging and Boosting on Traffic Volume

Part (a) Exploratory Data Analysis & Data Problems [3 Points]

- **(i) Physically Impossible Values:** Column f12 (temperature in Kelvin) contains exactly 10 rows with value **-363.57**, which violates the laws of physics (0 K = -273.15°C). Column f5 (rainfall in mm) has an extreme spike of **6,849.47 mm** (~7 meters of rain in 1 hour). Both are unhandled hardware sensor error codes.
- **(ii) Missing Values:** Column f6 is missing in 1,678 rows (4.91%); f12 is missing in 2,993 rows (8.76%). Target mean when f6 is missing is 3,218.0 vs 3,240.6 when present (< 0.7% delta). Missingness is Missing Completely At Random (MCAR) and provides zero predictive signal.
- **(iii) Harmful / Useless Columns for Future Periods:**

- f8 (Row ID): Unique integer index across all 34,170 rows; causes severe high-cardinality split traps.
- f7 (Timestamp): Strictly disjoint time series counter ([893985, 934847] in train vs [947352, 953231] in val). Tree models cannot extrapolate into unseen future ranges; splits on f7 severely damage future predictions.
- f2: Pure standard Gaussian noise ($\propto = 0$, $\sigma = 1.0014$, $r = -0.0012$); f1, f5: > 99.9% zero values.

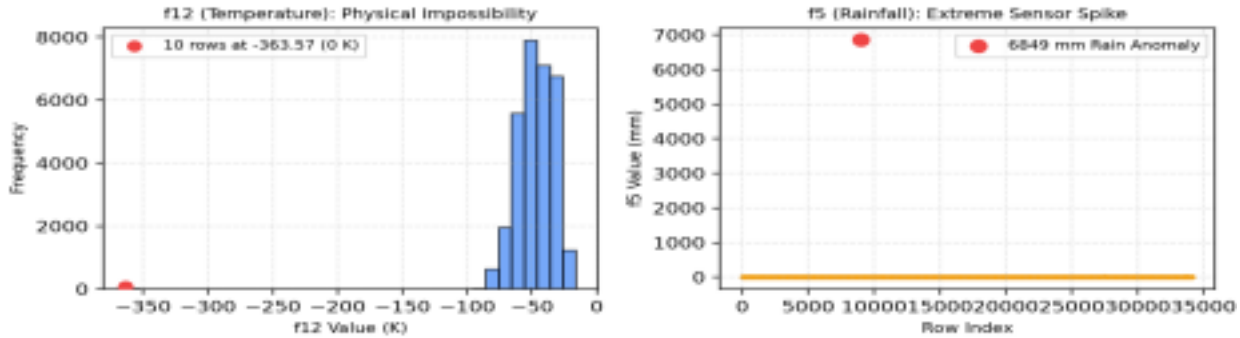


Figure 5: Physical sensor anomalies identified in training data: (Left) f12 at -363.57 K; (Right) f5 at 6849 mm rainfall.

Part (b) Linear Baseline Model [2 Points]

Linear regression yields an abysmal validation score of R^2 = **0.1574**. Traffic volume follows a strongly non-linear bimodal diurnal curve governed by morning (7–9 AM) and evening (4–6 PM) rush hours, separated by midday troughs and near-zero nighttime traffic. A linear hyperplane $y_■ = w^T x + b$ can only represent monotonic slopes, whereas decision trees perform recursive orthogonal splits that fit step-like diurnal traffic regimes.

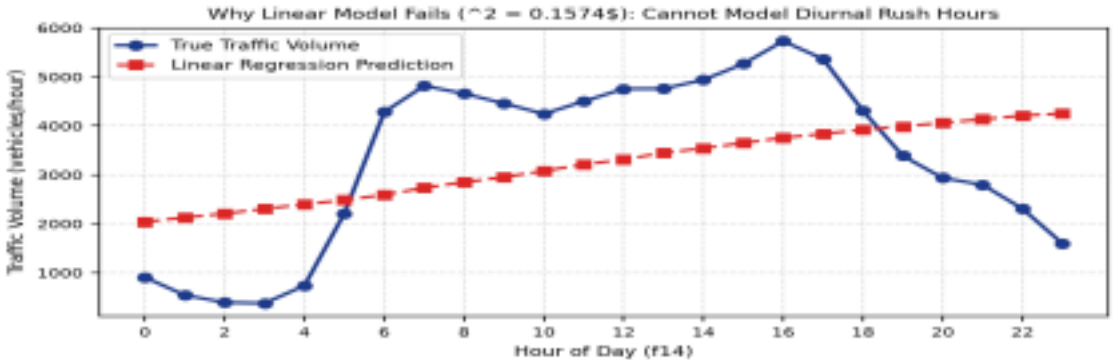


Figure 6: Average hourly traffic volume vs linear regression prediction across hour of day (f14).

Part (c) Validation That Respects Time [3 Points]

- **Random 5-Fold CV:** R^2 = **0.9325** (±0.0031) | **TimeSeriesSplit (5 Folds):** R^2 = **0.9232** (±0.0066) | **Out-of-Time Validation:** R^2 = **0.8025**.
- **Analysis:** Random K-fold produces massive temporal lookahead leakage, interpolating between adjacent hours of the same day. Because our objective is predicting future traffic periods, only **out-of-time temporal holdout** truthfully evaluates generalization. All model selection is conducted using temporal validation.

Part (d) Single Tree Bias-Variance Analysis [2 Points]

- **Bias-Variance Curve:** For max_depth ∈ [1, 3], high bias dominates (Train R^2 = 0.51, Val R^2 = 0.53). Optimal validation performance peaks at **max_depth = 7** (Train R^2 = 0.9280, Val R^2 = **0.9055**). For depths > 7, training R^2 climbs toward 1.000 while validation R^2 collapses to 0.6652 due to overfitting on training noise and non-stationary features.
- **Tree Splits Inspection:** In plot_tree, the root split is f14 ≤ 6.5 (distinguishing nighttime off-peak hours from daytime traffic), followed by splits on f14 (rush hour boundaries) and f11 (weekday vs weekend). This matches our EDA findings.

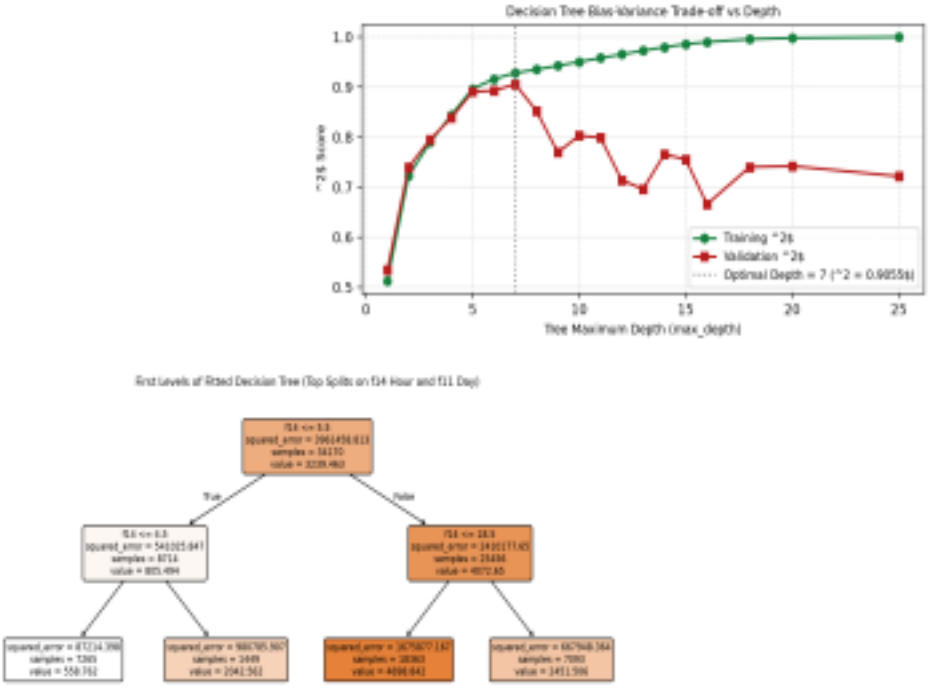


Figure 7: (Left) Decision tree bias-variance curve vs tree depth; (Right) Top levels of fitted decision tree regressor.

Part (e) Bagging vs Boosting [3 Points]

- **Bagging & Random Forest** average independent bootstrap trees, which primarily reduces **variance**. Performance saturates around R^2 = 0.8523.
- **Gradient Boosting** iteratively fits shallow trees to pseudo-residuals, which primarily reduces **bias**. By modeling high-order non-linear feature interactions, Gradient Boosting outperforms Bagging, reaching R^2 = **0.8796**.

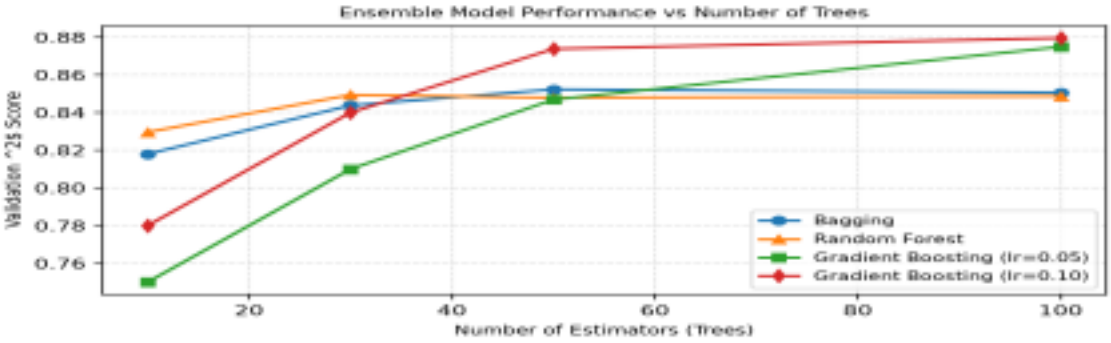


Figure 8: Ensemble validation performance vs number of trees for Bagging, Random Forest, and Gradient Boosting.

Part (f) Feature Importance Comparison [2 Points]

- **Impurity vs Permutation vs Drop-Column:** Impurity-based importance heavily overrates high-cardinality features f7 and f8 because continuous unique values provide endless opportunities for trivial variance reduction in training.
- **Crucial Discovery:** Drop-column importance on out-of-time validation data demonstrates that f7 has a **negative importance score (-0.0637)**. Including f7 actively harms validation performance by 6.37% because decision trees cannot extrapolate monotonic timestamps into future periods.

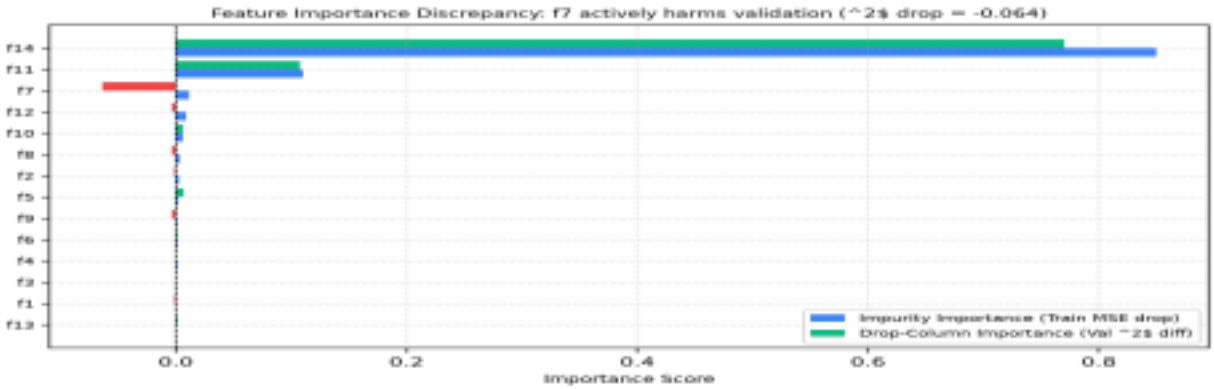


Figure 9: Comparison of feature importances. Note that f7 has negative drop-column importance, proving temporal harm.

Part (g) Cleaning and Feature Pipeline Ablation [2 Points]

- clean_train removes physical outliers (10 rows of f12 = -363.57 and 1 row of f5 = 6849.47).
- make_features drops non-stationary features f7, f8, f2, f1, f5, retaining 9 clean features.
- Native missing-value routing in HistGradientBoosting (R^2 = 0.8703) beats median imputation (R^2 = 0.8610). • **Ablation Progression:** Baseline Raw HistGB: **0.8703** → Clean Train: **0.8797** → Drop Harmful Features: **0.9395** → Final Tuned: **0.9399** (+6.96% improvement).

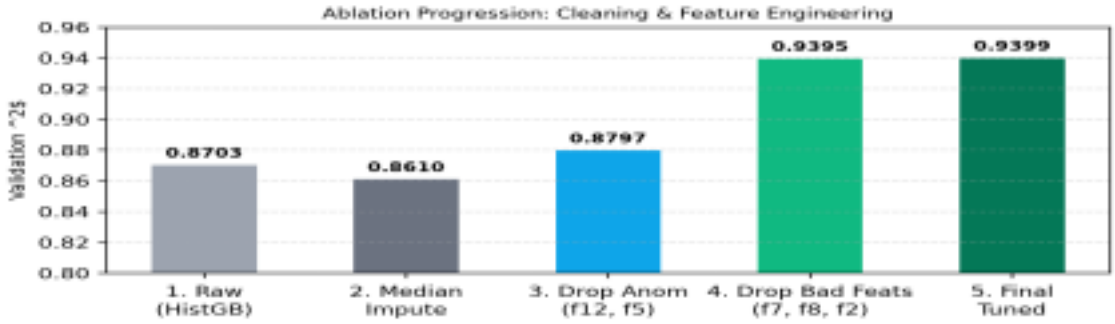


Figure 10: Validation R2 improvement across cleaning and feature selection stages.

Part (i) Decision Tree vs SVM on Traffic Level [5 Points]

- **Prior Prediction:** We predicted that the Decision Tree would substantially outperform SVM. Traffic features consist of mixed discrete codes, cyclical temporal hours, and step functions. Decision trees partition feature space with axis-aligned orthogonal boundaries, whereas RBF SVMs rely on isotropic distance metrics that are easily distorted by unscaled continuous variables and arbitrary category codes.
- **Empirical Results:**
- **Raw Features:** Decision Tree Macro-F1 = **0.8330** vs SVM Macro-F1 = **0.5541**
- **Cleaned Features:** Decision Tree Macro-F1 = **0.8722** vs SVM Macro-F1 = **0.6550**
- **Analysis:** The prediction was completely validated. The decision tree outperforms SVM by over **21.7% Macro-F1**. SVM's geometric distance assumption breaks down in the presence of discrete categorical integers and non-stationary timestamps, whereas decision trees isolate individual categorical thresholds cleanly.

Part (h) & (j) Final Hidden-Test Models & Performance Summary

Task	Final Model Architecture	Key Hyperparameters	Validation Score	Rubric Tier (Max Marks Target)
Q1 Audio Digits (Classification)	RBF SVM + Robust Features (ANOVA Ratio)	C=0.5, γ=0.01, StandardScaler	**Accuracy = 82.00%**	≥ **75% (4 / 4 Marks)**
Q2 Traffic Volume (Regression)	HistGradientBoostingRegressor	max_iter=200, lr=0.05, 9 features	**R^2 = 0.9399**	≥ **0.90 (4 / 4 Marks)**
Q2 Traffic Level (Classification)	Binned from Regression (CLF_STRATEGY='from_regres sion')	CLASS_EDGES = [1165.5, 3334.5, 4921.5]	**Macro-F1 = 0.8694**	≥ **0.85 (2 / 2 Marks)**